A Comprehensive Analysis of Sleeping Patterns and Disorders Among US Adults, 2017–2020

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1 Project Scope

1.1 Introduction

Sleep, like food, water, and oxygen, is essential to human life. The timing of sleep is influenced by biological factors, as evidenced by the human tendency to sleep at night. However, many factors, including education, occupation, social activities, insomnia, and various sleep and medical disorders, can result in individuals sleeping at varying times (Grandner, 2017, p. 1). The internet has significantly changed people's personal and professional lives, directly or indirectly influencing the sleep cycle, such as accessing social media in bed or browsing late at night (Grandner, 2017, p. 2).

Circadian rhythms and sleep habits are currently being altered in individuals. Due to various causes, they rely excessively on weekdays and weekends, negatively affecting metabolic function and increasing the risk of many diseases. Because of the differing sleep cycles on weekends and weekdays, a substantial percentage of sleep debt is compensated on weekends. Social jetlag is the difference between weekends and weekdays, as well as between social time and circadian rhythm (Roenneberg et al., 2012, p. 939). For instance, the prevalence of adults who sleep less than 6 hours may have increased over time in the United States, according to the minimal statistics available (Ford et al., 2015, p. 829). Finding out the usual time people go to sleep and wake up during the week and on weekends can help healthcare professionals understand sleep patterns and how they affect health. This can help promote healthy sleep habits and improve the health of people in the US.

1.2 Aim

The aim of this study is to find out how the people of the United States sleep and to evaluate the following things:

1. This study will look at how long people sleep on average, when they wake up on average, and how long people sleep on average on weekdays and weekends.
2. Determine the amount of sleep debt and social jetlag.
3. This study aims to investigate whether individuals who maintain a consistent sleep pattern throughout the week, including weekends, experience reduced daytime sleepiness.
4. This study aims to establish a correlation between sleep duration during weekdays and weekends and various factors such as snoring and snorting.

1.3 Purpose

This project aims to understand the US population's sleeping habits and sleep disturbances. With further research, these findings could be used to assist healthcare professionals understand sleep patterns and their effects on promoting healthy sleep habits and improving the health of US citizens. The results of this study can be used as a basis to investigate and improve the sleep habits of the US population.

1.4 Hypotheses

1. Determining if the average sleep-wake time is different on weekends and weekdays.
   Null hypothesis: The average sleep-wake time is not different on weekends and weekdays.
   Alternate Hypothesis: The average sleep and wake times differ on weekdays and weekends.

2. Determining if individuals who have regular sleep cycle is associated with lower daytime sleepiness.
   Null Hypothesis: Individuals who have regular sleep cycle is not associated with lower daytime sleepiness.
   Alternate Hypothesis: Individuals who have regular sleep cycle is associated with lower daytime sleepiness.

3. Determining if sleep duration on weekdays and weekends is associated with snoring, snorting, trouble sleeping, or feeling over-sleepy during the day.
   Null Hypothesis: Sleep duration on weekdays and weekends has no association with snoring, snorting, difficulty sleeping, or feeling sleepy during the day.
   Alternate Hypothesis: Sleep duration on weekdays and weekends has association with snoring, snorting, difficulty sleeping, or feeling sleepy during the day.

2 Data Exploration

2.1 Data collection and extraction

The National Health and Nutrition Examination Survey (NHANES) is an annual survey that gathers information on a wide range of health and nutritional issues from a nationally representative sample of the population in the United States of America. The Sleep Disorders Dataset (2017-20) comprises 10,196 rows and 11 columns. This data collection contains questions derived from the Munich Chrono Type Questionnaire about sleep patterns and problems, such as regular sleep times on weekdays and weekends, Sleep hours, Snorting, difficulty sleeping, and feeling drowsy during the day. Participants were surveyed at home using the Computer-Assisted Personal Interview (CAPI) technology. The precision and comprehensiveness of the data ensured quality assurance. Data set Link

Table 1: Sleep Disorders Dataset

SEQN	SLQ300	SLQ310	SLD012	SLQ320	SLQ330	SLD013	SLQ030	SLQ040	SLQ050	SLQ120
109266	22:00	05:30	7.5	23:00	07:00	8.0	1	0	2	0
109267	00:00	08:00	8.0	03:00	11:00	8.0	0	0	2	2
109268	22:00	06:30	8.5	23:00	07:00	8.0	0	0	2	1
109271	23:00	09:00	10.0	23:00	12:00	13.0	0	0	1	3
109273	08:00	14:35	6.5	21:00	05:00	8.0	0	0	1	2
109274	21:30	07:00	9.5	21:30	07:00	9.5	1	0	2	0

The data cleaning and data analysis were done using RStudio. There were 0.39% missing values in the dataset. According to the NHANES, missing values should only be imputed if they make up more than 10%

of the total observations. In this case, less than 10% of the observations were missing values, so they were not imputed. In this project, the observations with missing data were omitted as they were in important variables like sleep time, wake time, and sleep disturbances, which would have been useless if they had been kept with the missing data and considered only the complete data for the important variables. The column headers were modified to enhance the understanding of the variables.

2.2 Variables

Variables include a Sequence number, weekdays sleep time, weekdays waketime, weekdays sleep time, weekends sleep time, weekends wake time, weekends sleep duration, Snoring frequency, Snorting, Trouble sleeping, and Excessive sleep during the day which are coded as SLQ and SLD. Sequence number is discrete, snoring, Snorting and oversleepy during the day are ordinal variables, Trouble sleeping is a binary variable, Sleep time, wake time on weekends and week days are in the time format and are continuous. Sleep duration on weekends and weekdays are continuous variables. In addition other variables like Social jetlag (mid-point d ifference between sleep on weekdays and weekends) and sleep debt (the difference between weekend sleep duration and weekly average sleep duration) will be determined (Di et al., 2022, p. 2).

2.3 Data Distribution and Summary

The summary statistics in Table 2 provide an overview of sleeping patterns. Important variables include the duration of sleep on weekdays `WD_sleephours` and weekends `WE_sleephours`. The minimum weekday and weekend sleep duration is 3 hours, and the maximum is 14 hours. The average weekday sleep duration is 7.65 hours, with a median of 7.5 hours. The interquartile range (IQR) is between 7 and 8.5 hours, representing the middle 50 percent of the data. The average weekend sleep duration is 8.36 hours, with the median being 8 hours as shown in Figure 2. The IQR ranges between 7 and 9.5 hours, representing the median 50% of the data set. The `Snoring` and `Snorting` variables ranges from 0 (Never) to 9 (Don't know), with a median of 1 (Rarely) and a median of 0 (Never) simultaneously and they both are right skewed (See Appendix). The median value of the trouble sleeping variable `trouble_sleep` is (No), ranging from 1 (Yes) to 9 (Don't know). Over-sleepy during the day variable `Oversleepy` ranges from 0 (Never) to 9 (Don't know), with a mean of 2 (Sometimes) and is rightly skewed. The average `sleep_debt` is 0.50 hours. Absolute sleep debt was calculated on weekends and weekdays by taking 8 hours as the recommended sleep duration as shown in Figure 2.

Table 2: Descriptive Statistics

	WD_sleephours	WE_sleephours	Snoring	Snorting	Trouble_sleep	Oversleepy	sleep_debt
Min.	3.000000	3.000000	0.000000	0.000000	1.000000	0.000000	-9.000000
1st Qu.	7.000000	7.000000	0.000000	0.000000	1.000000	1.000000	0.000000
Median	7.500000	8.000000	1.000000	0.000000	2.000000	2.000000	0.000000
Mean	7.653565	8.358017	1.939035	0.820014	1.739096	1.753033	0.704452
3rd Qu.	8.500000	9.500000	3.000000	1.000000	2.000000	3.000000	1.500000
Max.	14.000000	14.000000	9.000000	9.000000	9.000000	9.000000	10.000000

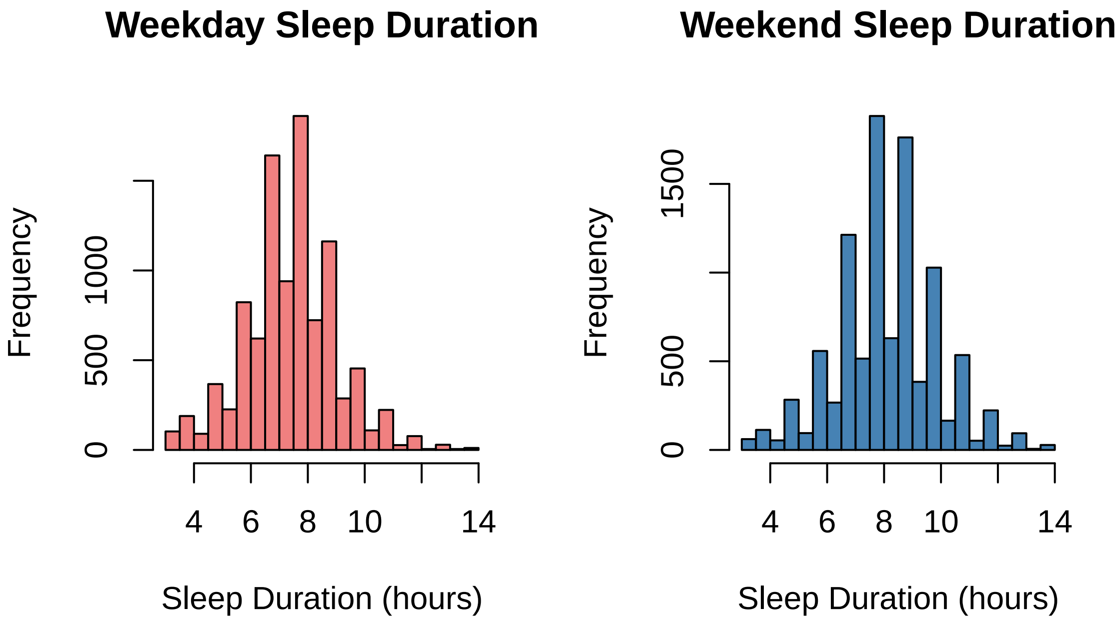


Figure 1: Distribution of sleep hours

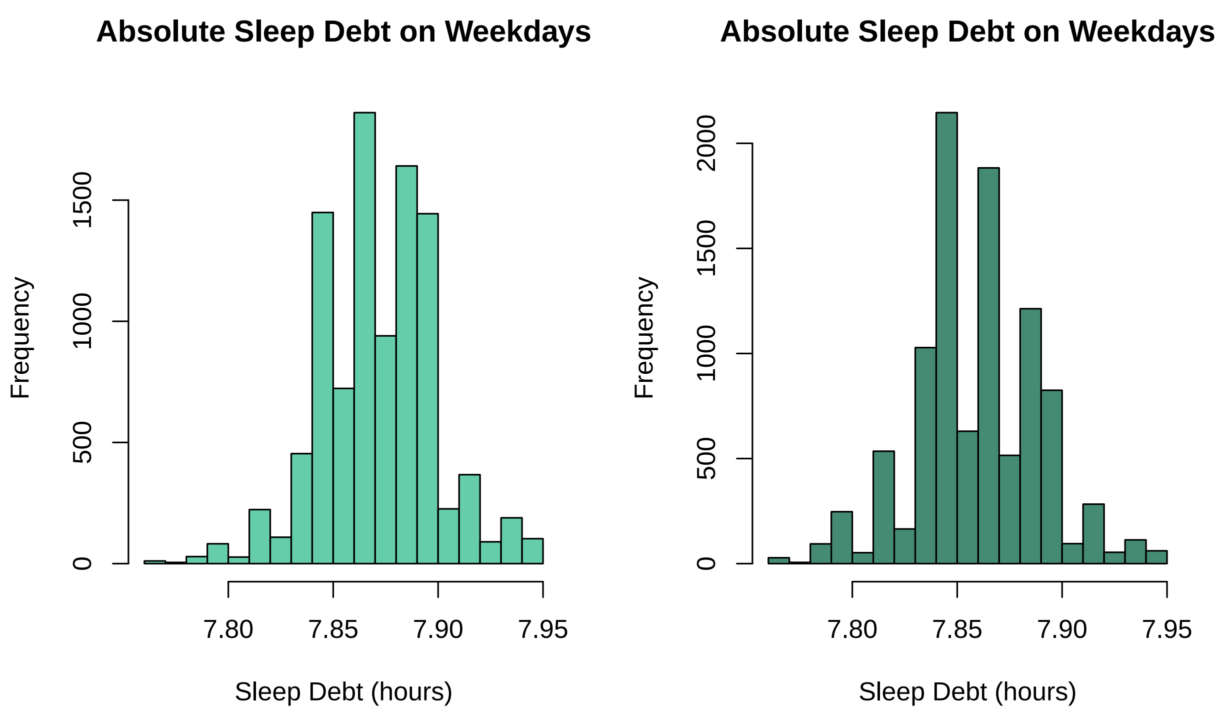


Figure 2: Sleep debt compared to recommended 8 hours of sleep

3 Statistical Methods

Exploratory Data Analysis

Individuals who wake up at the same time on weekdays and weekends and get the same amount of sleep on both days are investigated to see if a regular sleep cycle reduces daytime sleepiness. To find the correlation, the ANOVA will compare the groups with and without a healthy sleep cycle. Given that the assumption of homogeneity of variance is not met for variance analysis, the Kruskal-Wallis test is utilized as a non-parametric alternative. The Kruskal-Wallis test indicates no significant difference in daytime `Oversleepy` between those with and without regular sleep cycles (p-value = 0.6854). Therefore, maintaining a regular sleep cycle does not significantly impact daytime sleepiness. As the chi-square is less for Kruskal-Wallis this forms one of the limitations of this statistical test.

To determine the multiple linear regression, log transformation was done on `Snoring,` and multiple linear regression was done using variables like sleep debt, trouble sleeping, and over-sleepiness as multicollinearity was determined for them. The adjusted R square was determined by testing different models, and the adjusted r square for `WD_sleephours` was 18%, and for`WE_sleephours`, it was 30%, determining the best fit for the model. Outliers were also determined, but as the project deals with self-reported data, the outliers were not excluded from the dataset as they might also give important information. Robust regression was done to address the presence of outliers or influential points. For the robust regression, the residual error was less when compared to multiple linear regression,but the adjusted R square was low when compared to multiple linear models. One of the limitations of this is the adjusted R square was less than the models. The normal Q-Q plots were plotted (See Figure 3), which slightly deviated from the normal, but as the population size is large, this was not considered.

For the logistic regression, a correlation heatmap was plotted to determine the highly associated factors. For the logistic regression, the binary variable `Trouble_sleep` was plotted against other independent variables, and the most association was found between `Oversleepy`, `Snoring`, and `Snorting`. The multiple logistic regression's independent variables were all categorical, so they were converted to factors to yield a best-fitting model. For this AIC value was 10726, which was best fitting when compared to other models.

Limitations encountered during the project

1. As the data was self-reported, the outliers could not be removed, which might have significantly affected the data.
2. As the data was in time series, it was difficult to determine the exact sleep times as the meantime could not be exactly determined.
3. As the data were not normally distributed, ANOVA could not be tested to determine the mean across groups, and Kruskal-Wallis yielded a low chi-square value.
4. The multiple linear regression yielded less adjusted R square for `WD_sleephours`, which might indicate certain missing data or other factors which are not associated with it.
5. The multiple logistic regression yielded high AIC due to the large sample size.

4 Results

In the US population from 2017 to 2021, the mean sleep duration on weekdays was 7.65 hours (95% CI, 7.62-7.69 hours). The mean sleep duration on weekends was 8.36 hours (95% CI, 8.32-8.40 hours). The difference between weekday and weekend sleep duration was 0.70 hours. It indicated that individuals slept more on weekends than on weekdays. Sleep and wake times were also differed among the US population. The mean wake time on weekdays was 06:44:51 hours (95% CI, 06:42:10- 06:42:10 hours). The mean wake time on weekends was 07:52:34 hours (95% CI, 07:50:01-07:55:08 hours). The difference between

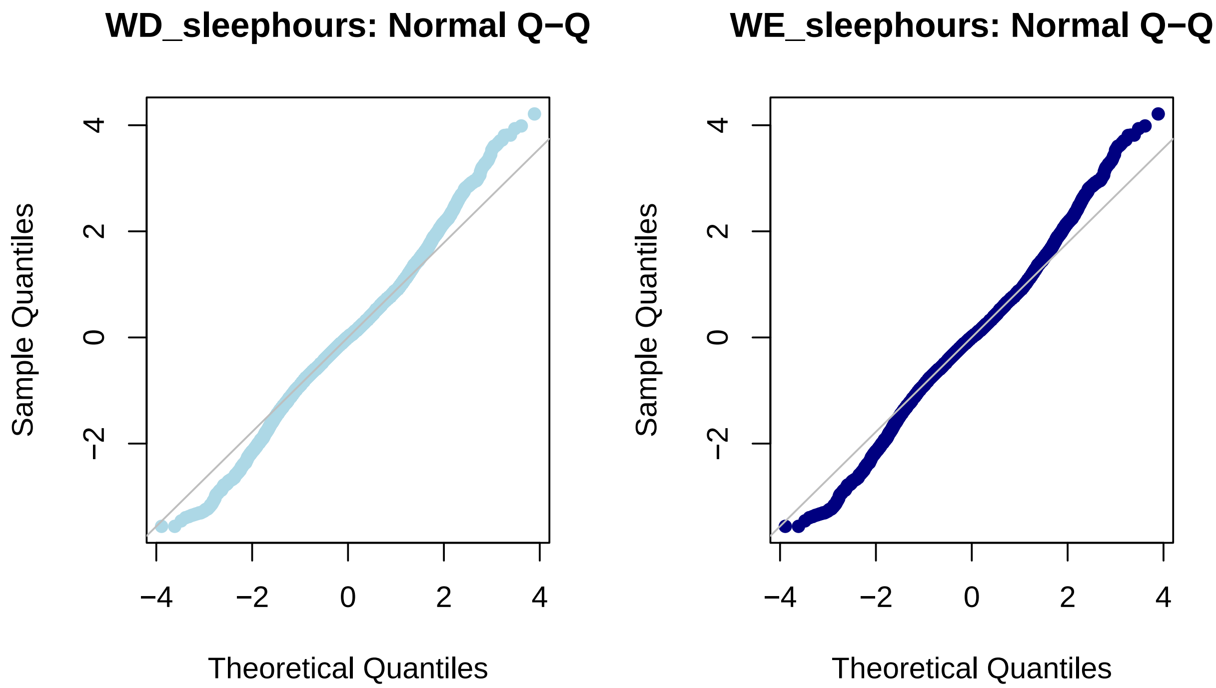


Figure 3: Normal Q-Q plots

wake times on weekdays and weekends was 01:07:43 hours.The mean sleep time on weekends and weekdays is 12:00 AM. The mean sleep debt was 0.50 hours (95% CI, 0.48-0.53 hours). The Kruskal-Wallis test was performed to determine whether individuals with regular sleep cycles also experience excessive daytime sleepiness. The p-value of 0.6851 indicates that there was no statistically significant difference in the distribution of Oversleepy scores between those with regular sleep cycles and those without, so failing to reject the null hypothesis where regular sleep cycle is [regular_sleep_cycle = (Weekday_waketime == Weekend_waketime) & (Weekday_sleephours == Weekend_sleephours)].

The multiple linear regression was done on between the dependent variable of weekday sleep hours and several independent variables, including log-transformed snoring, sleep debt, trouble sleeping, and oversleepiness. The statistical analysis reveals that the model as a whole is statistically significant (F-statistic = 538.3, $p < 2.2e\text{-}16$). Moreover, the adjusted R-squared value suggests that the explanatory variables account for approximately 18% of the variability in the response variable. The regression analysis reveals that the variable of snoring, which has been subjected to log transformation, exhibits a negative coefficient of -0.17192. This suggests an inverse relationship exists between the level of snoring and the number of hours of sleep during weekdays, whereby an increase in snoring is associated with a decrease in weekday sleep hours. The findings reveal a negative correlation between sleep debt and weekday sleep hours, as evidenced by the negative coefficient value of -0.57194. This suggests that an increase in sleep debt is associated with a decrease in the number of hours slept during weekdays. The coefficient for trouble_sleep is positively significant (0.15174), implying that an increase in the difficulty level is associated with an increase in the number of hours of sleep during weekdays. The variable Oversleepy exhibits a negative coefficient value of -0.05041, which suggests an inverse relationship between the degree of over sleepiness and the duration of sleep hours on weekdays. Another multiple linear regression was done on Weekend sleephours and log-transformed snoring, sleep debt, trouble sleeping, and over-sleepiness which resulted in Multiple R-squared:0.3071, Adjusted R-squared: 0.3068 indicating a best fitting model when compared to others.

After performing the correlation for the binary variable `Trouble_sleep` to other variables like `Oversleepy`, `Snoring`, `Snorting`, `sleep_debt`, `WD_sleephours` and `WE_sleephours`. The correlation was found for `Trouble_sleep`, `Oversleepy`, `Snoring`, `Snorting` and multiple logistic regression was performed which yielded. The findings indicate that with the exception of Oversleepy9 and Snoring1, all predictor variables demonstrated statistical significance in predicting trouble_sleep. The study reported the odds ratio for each level of the predictor variables along with their corresponding p-values. The adequacy of the model fit was evaluated by means of the deviance residuals and the AIC value which is 10726. The adequacy of the model was demonstrated by the residual deviance being less than the null deviance, and the AIC value being comparatively low when compared to other models.The results indicate that trouble sleeping is associated with snoring, snorting, and excessive sleepiness during the day.

5 Strengths and Limitations of the project

One of the project's strengths is that it is based on a significant and diverse population. There are, however, several limitations to consider. Firstly, the study depends on self-reported sleep data, which is prone to memory bias because individuals may not remember their sleep periods precisely, resulting in misinterpretation of results. Second, missing data were eliminated from the study, which may have resulted in the loss of crucial information. However, keeping it may not have improved results in this context.

Furthermore, the study may not have considered all confounding factors that could influence sleep deprivation, such as profession, age, gender, and multiple disorders. Including these characteristics could provide a complete knowledge of people's sleeping habits. Finally, because the data format for sleep timings (hh: mm) may not accurately reflect the sleep duration or the precise sleep and wake times, it may have contributed to the misinterpretation of the results.

Despite these limitations, the study provides valuable information about sleep patterns and factors influencing sleep quality. Future studies could overcome these limitations by including more diverse factors, upgrading data collection methods, and analyzing the data using better statistical methods.

6 Conclusion

The project found significant disparities in sleep habits between weekdays and weekends among the US population. The results indicated that individuals tend to sleep longer on weekends than during the week, which is consistent with the notion of "sleep debt." Another important finding was the presence of social jet lag, which suggests that social jet lag is a prevalent phenomenon among the US population, with potentially detrimental effects on health and well-being. This research has significant implications for comprehending sleep patterns and their effects on health. Healthcare professionals can devise targeted interventions to promote healthy sleep habits and reduce the risk of sleep-related disorders by identifying weekday and weekend sleep differences. In addition, this project's findings can be used to investigate other factors that may contribute to sleep deprivation, such as shift work and stress, and to develop individualized strategies to improve sleep patterns among the US population.

References

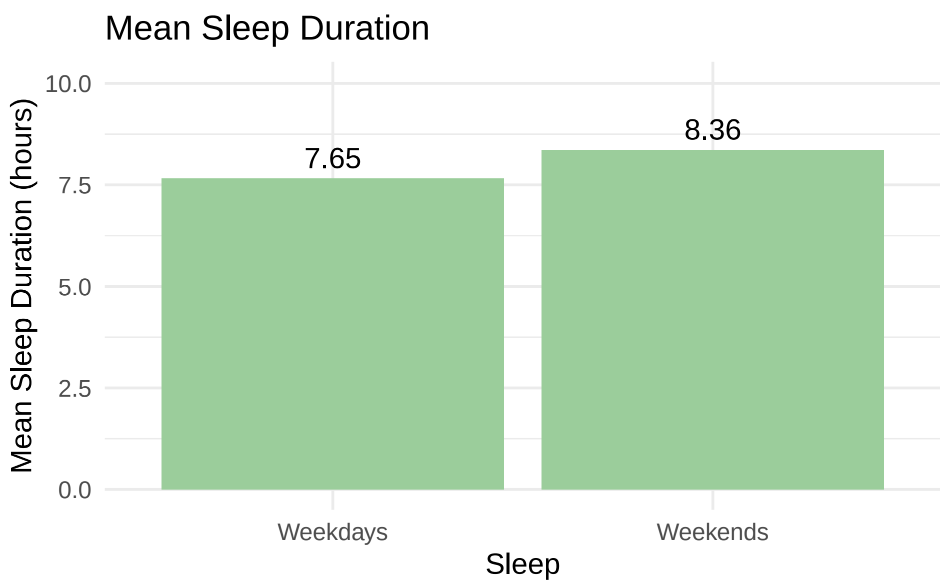
Di, H., Guo, Y., Daghlas, I., Wang, L., Liu, G., Pan, A., Liu, L., & Shan, Z. (2022). Evaluation of sleep habits and disturbances among US adults, 2017-2020. JAMA network open, 5(11), e2240788-e2240788. https://doi.org/10.1001/jamanetworkopen.2022.40788

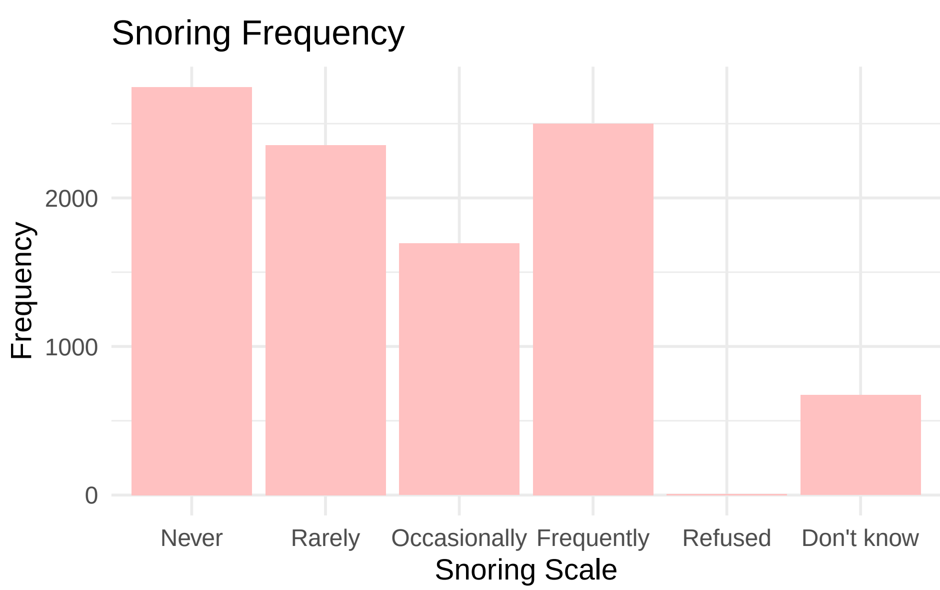
Ford, E. S., Cunningham, T. J., & Croft, J. B. (2015). Trends in self-reported sleep duration among US adults from 1985 to 2012. Sleep, 38(5), 829-832. https://doi.org/10.5665/sleep.4684

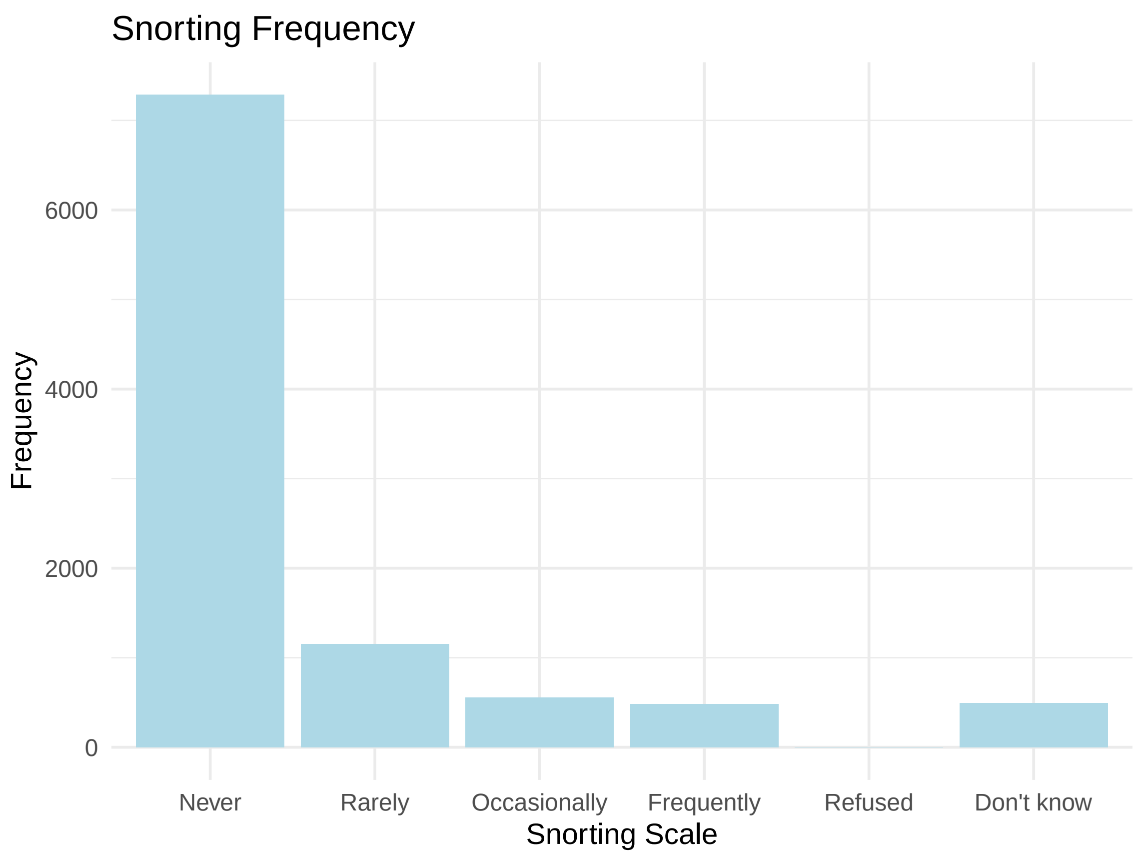
Grandner, M. A. (2017). Sleep, health, and society. Sleep medicine clinics, 12(1), 1-22. https://doi.org/10.1016/j.jsmc.2016.10.012

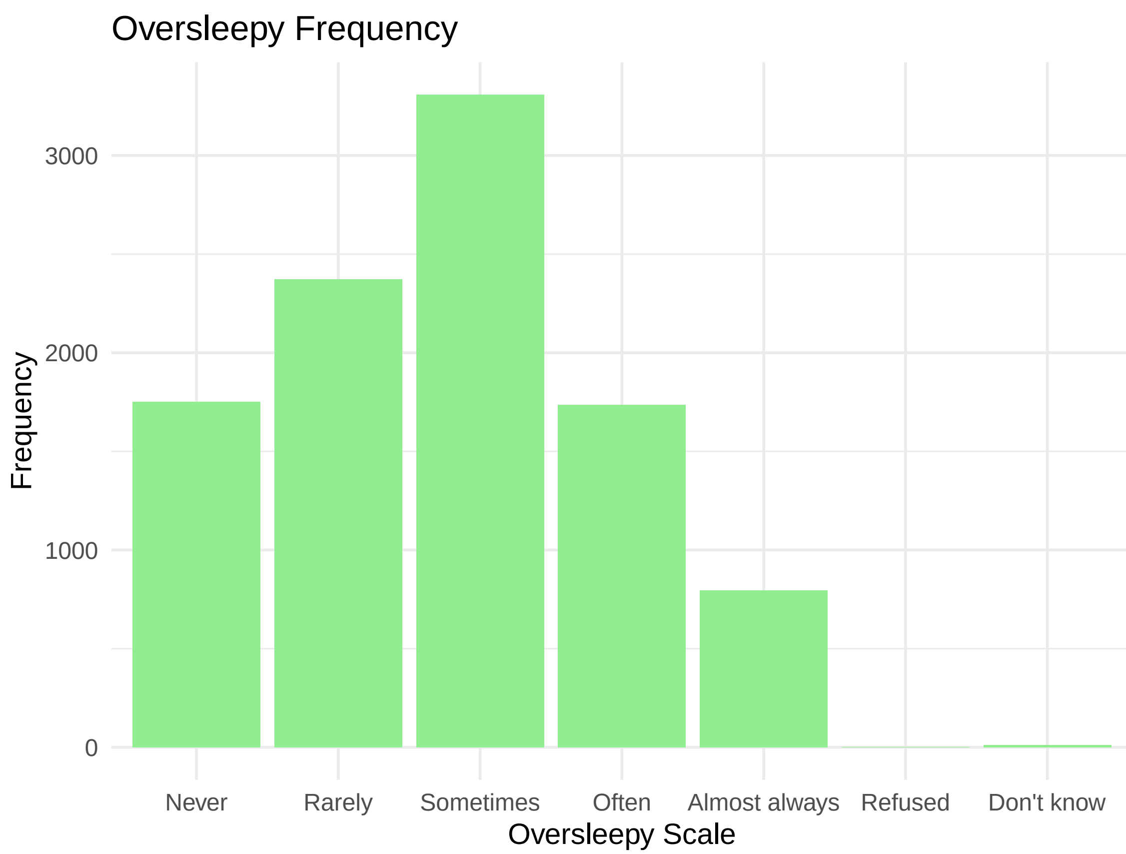
Roenneberg, T., Allebrandt, Karla V., Merrow, M., & Vetter, C. (2012). Social jetlag and obesity. Current Biology, 22(10), 939-943. https://doi.org/10.1016/j.cub.2012.03.038

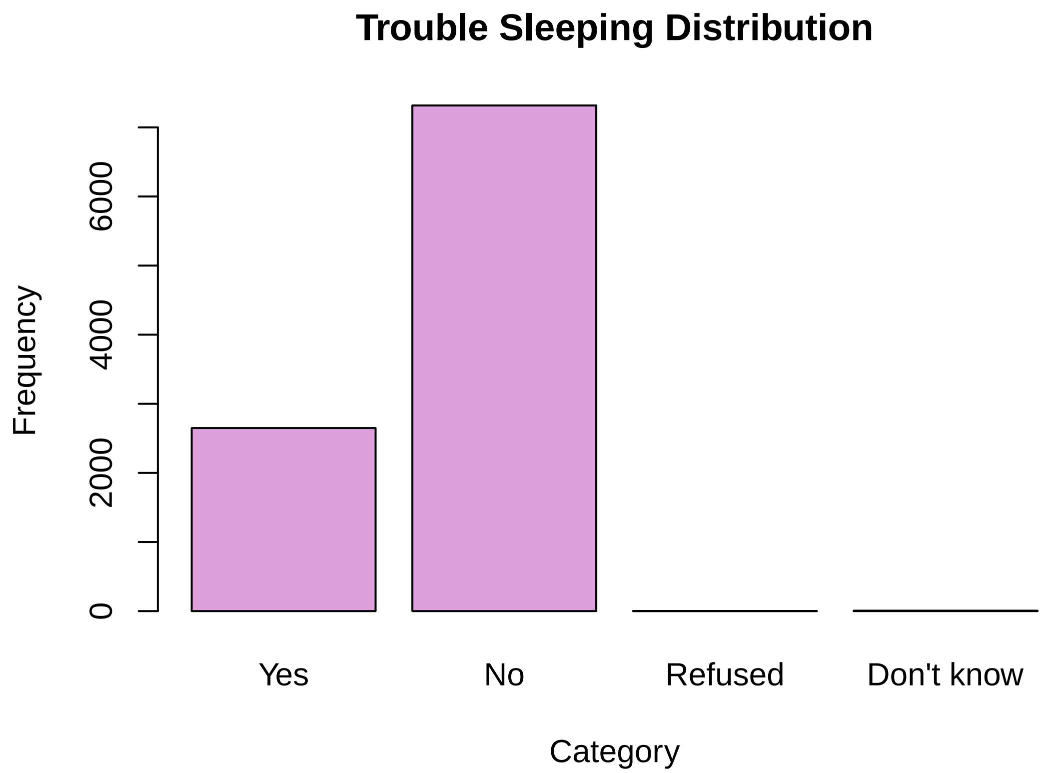
Appendix






Snorting Frequency
Frequency
6000
4000
2000
0
Never
Rarely
Occasionally
Frequently
Refused
Don't know
Snorting Scale


Oversleepy Frequency
Frequency
3000
2000
1000
0
Never
Rarely
Sometimes
Often
Almost always
Refused
Don't know
Oversleepy Scale

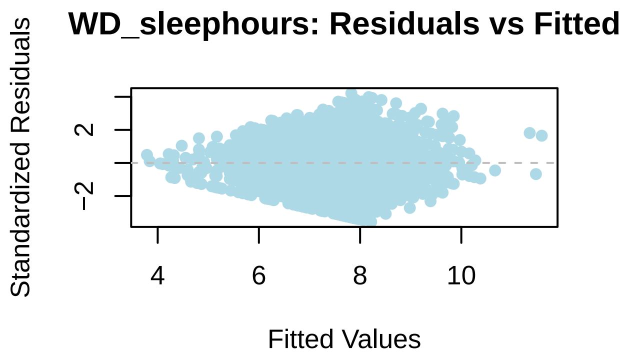


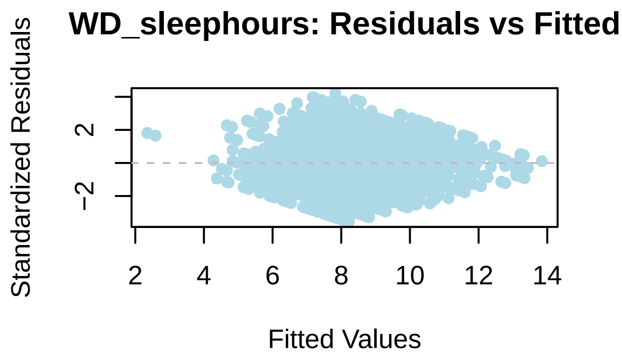

```
## kruskal_wallis test for regular sleep cycle and oversleepy:

##
##  Kruskal-Wallis rank sum test
##
## data:  Oversleepy by regular_sleep_cycle
## Kruskal-Wallis chi-squared = 0.1644, df = 1, p-value = 0.6851

## Multiple Linear Regression for weekdays:

##
## Call:
## lm(formula = WD_sleephours ~ log_Snoring + sleep_debt + Trouble_sleep +
##     Oversleepy, data = sleepdf)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -5.2162 -0.8761  0.0038  0.8846  6.1734
##
## Coefficients:
##                Estimate Std. Error t value Pr(>|t|)
## (Intercept)    7.912693   0.067812 116.686  < 2e-16 ***
## log_Snoring   -0.171919   0.022543  -7.626 2.64e-14 ***
## sleep_debt    -0.408531   0.009066 -45.062  < 2e-16 ***
## Trouble_sleep  0.151743   0.031410   4.831 1.38e-06 ***
```

```
## Oversleepy    -0.050406   0.012599  -4.001 6.36e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.466 on 9968 degrees of freedom
## Multiple R-squared:  0.1777, Adjusted R-squared:  0.1773
## F-statistic: 538.3 on 4 and 9968 DF,  p-value: < 2.2e-16
```

```
## Multiple Linear Regression for weekends:
```

```
##
## Call:
## lm(formula = WE_sleephours ~ log_Snoring + sleep_debt + Trouble_sleep +
##     Oversleepy, data = sleepdf)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -5.2162 -0.8761  0.0038  0.8846  6.1734
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept)    7.912693   0.067812 116.686  < 2e-16 ***
## log_Snoring   -0.171919   0.022543  -7.626 2.64e-14 ***
## sleep_debt     0.591469   0.009066  65.240  < 2e-16 ***
## Trouble_sleep  0.151743   0.031410   4.831 1.38e-06 ***
## Oversleepy    -0.050406   0.012599  -4.001 6.36e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.466 on 9968 degrees of freedom
## Multiple R-squared:  0.3071, Adjusted R-squared:  0.3068
## F-statistic:  1105 on 4 and 9968 DF,  p-value: < 2.2e-16
```






```
## Multiple Logistic Regression:

##
## Call:
## glm(formula = binary_trouble_sleep ~ Oversleepy + Snoring + Snorting,
##     family = binomial, data = clean_sleepdf)
##
## Deviance Residuals:
##     Min       1Q   Median       3Q      Max
## -1.6086  -0.7245  -0.6005   0.9184   2.0447
##
## Coefficients:
##             Estimate Std. Error z value Pr(>|z|)
## (Intercept) -1.95835    0.07750 -25.268  < 2e-16 ***
## Oversleepy1  0.22895    0.08633   2.652   0.0080 **
## Oversleepy2  0.57987    0.07928   7.314 2.59e-13 ***
## Oversleepy3  1.22884    0.08453  14.537  < 2e-16 ***
## Oversleepy4  1.52344    0.10011  15.218  < 2e-16 ***
## Oversleepy9 -0.05315    0.79188  -0.067   0.9465
## Snoring1     0.06897    0.07045   0.979   0.3276
## Snoring2     0.14152    0.07621   1.857   0.0633 .
## Snoring3     0.10792    0.07258   1.487   0.1370
## Snoring7     1.10326    0.87993   1.254   0.2099
## Snoring9     0.27092    0.10765   2.517   0.0118 *
## Snorting1    0.45667    0.07253   6.296 3.05e-10 ***
```

```
## Snorting2    0.77899    0.09696   8.034 9.44e-16 ***
## Snorting3    1.26670    0.10718  11.819  < 2e-16 ***
## Snorting7   -0.96226    1.40576  -0.685   0.4937
## Snorting9    0.59805    0.10949   5.462 4.70e-08 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 11541  on 9965  degrees of freedom
## Residual deviance: 10694  on 9950  degrees of freedom
## AIC: 10726
##
## Number of Fisher Scoring iterations: 4
```